AIMS African Institute for Mathematical Sciences RWANDA

Introduction to Probability and Statistics

Lecture 4: Probability IV

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1. Moments

2. Bivariate random variables

3. Independent r.v.'s

Examples of discrete r.v.'s distribution

Definition

A r.v. X is said to be a ***Bernoulli****'s r.v. with parameter p iff*

$$P(X = x) = p^x (1-p)^{1-x}, \ x \in \{0, 1\}, \ 0 < p < 1.$$

Notation: $X \sim \mathscr{B}(1, p)$.

Definition

A r.v. X follows a ***binomiale*** *distribution with parameters (n, p) iff*

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} \ , \ x \in \{0, \cdots, n\} \ 0 < p < 1.$$

Notation: $X \sim \mathscr{B}(n, p)$.

Expectation

Definition

*The **expected value** or **mean** or **first moment** of a random variable X is defined as:*

$$E(X) = \begin{cases} \displaystyle\sum_{x \in V} x \; P(X = x) & \textit{if } X \textit{ is discrete,} \\[2em] \displaystyle\int_V x \; f_X(x) dx & \textit{if } X \textit{ is continuous.} \end{cases}$$

Example:

- $X \sim \mathscr{B}(1, p)$.
- $X \sim \mathscr{E}(\theta)$.

Moments

- Let g be a regular function. Then:

$$E\left[g(X)\right]) = \begin{cases} \displaystyle\sum_{x \in V} g(x)\ P(X = x) & \text{if } X \text{ is discrete} \\ \displaystyle\int_V g(x)\ f_X(x)dx & \text{if } X \text{ is continuous.} \end{cases}$$

- q^{th}-moments

$$E(X^q) = \begin{cases} \displaystyle\sum_{x \in V} x^q\ P(X = x) & \text{if } X \text{ is discrete,} \\ \displaystyle\int_V x^q\ f_X(x)dx & \text{if } X \text{ is continuous.} \end{cases}$$

m.g.f.

Moment generating function (m.g.f)

$$M_X(t) = E\left[e^{tX}\right].$$

Lemma

$$E(X^q) = M_X^{(q)}(0).$$

➥ The m.g.f identifies the distribution.

Expectation

Properties

- $\forall b \in \mathbb{R},\ E(X+b) = E(X) + b.$
- $\forall a \in \mathbb{R},\ E(aX) = aEX.$

Exercise

A fair coin is tossed n times. Let X be the number of TAIL.
X is a Binomiale distribution with parameters (n, p) with $p = 1/2$

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \ldots, n.$$

Let a be a real number. Consider the random variable $Y = \dfrac{a^X}{2^n}$.
Compute $E(Y)$.

Expectation

Solution:

By definition, $E(a^X/2^n) = \sum_{x=0}^{n} (a^X/2^n) P(X=x)$.

Hence

$$\begin{aligned} E(Y) &= \frac{1}{2^n} \sum_{x=0}^{n} a^x \binom{n}{x} \left(\frac{1}{2}\right)^x \left(1 - \frac{1}{2}\right)^{n-x} \\ &= \frac{1}{2^n} \sum_{x=0}^{n} a^x \binom{n}{x} \left(\frac{1}{2}\right)^n = \frac{1}{2^{2n}} \sum_{x=0}^{n} \binom{n}{x} a^x = \left(\frac{a+1}{4}\right)^n \end{aligned}$$

Reminder:

Newton's binomial formula : $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$.

Variance of a random variable

Definition

The **variance** *of a random variable is defined as:*

$$Var(X) = E\left[(X - E(X))^2\right].$$

$$Var(X) = \begin{cases} \sum_{x \in V} \left(x - E(X)\right)^2 P(X = x) & \text{if } X \text{ is discrete} \\ \int_V \left(x - E(X)\right)^2 f_X(x) dx & \text{if } X \text{ is continuous.} \end{cases}$$

Variance of a random variable

Exercise: Compute the variance for:

- $X \sim \mathscr{B}(1, p)$.
- $X \sim \mathscr{E}(\theta)$.

Variance of a random variable

Properties

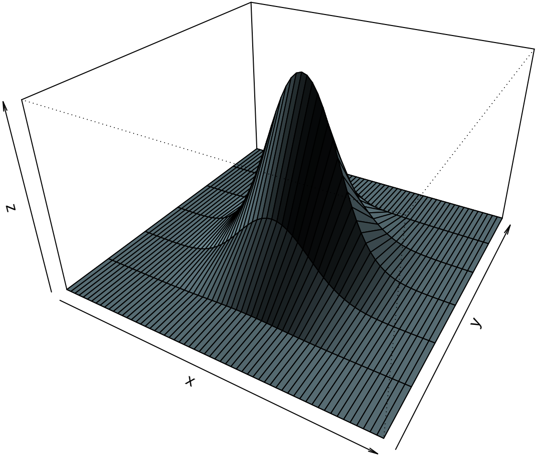
- $Var(X) = E(X^2) - (EX)^2$.
- $\forall b \in \mathbb{R},\ Var(X+b) = Var(X)$.
- The variance is **quadratic** : $\forall a,\ Var(aX) = a^2 Var(X)$.

Couple of r.v.'s

- Distribution of a bivariate random variables (joint distribution): For $(x, y) \in U \times V$,
  - $P(X = x, Y = y)$, if X and Y are discrete r.v.'s.
  - $f_{X,Y}(x, y)$, if X and Y are continuous r.v.'s (bivariate p.d.f.)
- Conditions:
  - $\sum_{(x,y)} P(X = x, Y = y) = 1$ (discrete),
  - $\int_{U \times V} f_{(X,Y)}(x, y) dxdy = 1$ (continuous).

Bivariate random variable

Example of a bivariate r.v. representation



- **Marginale distributions**
  - $P(X=x)=\sum_y P(X=x,Y=y)$ and $P(Y=y)=\sum_x P(X=x,Y=y)$ (discrete)
  - $f_X(x)=\int f_{(X,Y)}(x,y)dy$ and $f_Y(y)=\int f_{(X,Y)}(x,y)dx$ (continuous).
- **Conditional distribution**
  - $P(X=x \mid Y=y)=\dfrac{P(X=x,Y=y)}{P(Y=y)}$ (discrete)
  - $f_{X|Y=y}(x)=\dfrac{f_{(X,Y)}(x,y)}{f_Y(y)}$ (Continuous version of the Bayes's theorem).

Expectation and Covariance

- $$E\left[g(X,Y)\right] = \begin{cases} \sum_{x,y} g(x,y)P(X=x, Y=y) & \textit{if } X \textit{ and } Y \textit{ are discrete,} \\ \int g(x,y) f_{X,Y}(x,y)dxdy & \textit{if } X \textit{ and } Y \textit{ are continuous.} \end{cases}$$

Expectation and Covariance

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- $E(X+Y) = EX + EY$

Expectation and Covariance

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Expectation and Covariance

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- $E(X+Y) = EX + EY$
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- The **covariance** of two random variables is:

$$Cov(X,Y) = E\left[(X-EX)(Y-EY)\right] = E(XY) - E(X)E(Y).$$

Expectation and Covariance

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- The **covariance** of two random variables is:
  $$Cov(X,Y) = E\left[(X-EX)(Y-EY)\right] = E(XY) - E(X)E(Y).$$
- **Remark:**
  $Cov(X,X) = E\left[(X-EX)(X-EX)\right] = E\left[(X-EX)^2\right] = Var(X).$

Independent r.v.'s

Definition

Two random variables are said to be independent if the distribution of the couple is the product of each variable distribution.

$$(X \text{ and } Y \text{ independent}) \Longleftrightarrow \begin{cases} P(X=x, Y=y) = P(X=x)P(Y=y), \\ \text{for discrete r.v.'s.} \\ \\ f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y), \\ \text{for continuous r.v.'s.} \end{cases}$$

Independent r.v.'s

Exercise – Let $f_{X,Y}(x,y) = 4(1-x)y$, for $x, y \in [0,1]$ and $f_{X,Y}(x,y) = 0$ otherwise.

1. Show that $f(x,y)$ is a joint probability distribution function.
2. Find the marginal p.d.f. of X and Y.
3. Are X and Y independent?
4. Find the probability $P(X \leqslant 1/2 \mid Y > 1/3)$.

Independent r.v.'s

Proposition

Let X and Y be two r.v.'s
If X and Y are independent then $Cov(X, Y) = 0$.

Remark: $Cov(X, Y) = 0 \Longleftrightarrow E(XY) = E(X)E(Y)$.

Warning: $Cov(X, Y) = 0$ does not imply that X, Y are independent.

Exercise:
Let (X, Y) be a bivariate r.v. taking values in $V = \{(-2,4),(-1,1),(0,0),(1,1),(2,4)\}$ such that $P(X = i, Y = j) = 1/5,\ \forall (i,j) \in V$.

1. Are X and Y independent?
2. Compute $Cov(X, Y)$.

Independent r.v.

Proposition

Let $X_1, X_2, \dots, X_n$ be n idependent r.v. Denote $S_n = \sum_{i=1}^{n} X_i$.

Then: $M_{S_n}(t) = \prod_{i=1}^{n} M_{X_i}(t)$.

Proof: $M_{S_n}(t) = E\left(e^{tS_n}\right) = E\left(e^{t\sum_{i=1}^{n} X_i}\right) = E\left(\prod_{i=1}^{n} e^{tX_i}\right)$.

Since the r.v. are independent: $E\left(\prod_{i=1}^{n} e^{tX_i}\right) = \prod_{i=1}^{n} E\left(e^{tX_i}\right)$.

Therefore $M_{S_n}(t) = \prod_{i=1}^{n} M_{X_i}(t)$.